

Course Code: CS 498

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Office Hours: M/W: 1:00pm-3:45pm

Discord: <https://discord.gg/..>

Office: On Discord

Course Description

Students who are enrolled in this class will be exposed to the basics of Penetration Testing. Also, students will practice documenting and reporting the findings. This is done to simulate real-world penetration testing environments.

This course will be taught asynchronously, meaning there will be no dedicated class time. All material will be self-guided. We will have weekly goals that are to be completed. These will be outlined in the schedule below. It is recommended to stay ahead of schedule and accomplish as much as possible while you can. Additional reading material about a topic will be provided upon request.

I encourage the use of AI to understand a topic at a deeper level, or for help with a lab. However, do not use AI as a crutch. The work you put into this course, you will get back in return. If you overuse AI you are robbing yourself from learning valuable skills, which you can add to your resume.

I will be on Discord M/W: 1:00pm-3:45pm if any students have questions about a lab. Feel free to reach out via Discord or email outside of the hours.

Required Materials

- TryHackMe.com Premium subscription will be provided
- Overleaf account needed for LaTeX editor
- Additional book material will be provided at request

Requirements

- Students should be comfortable using a Command Line Interface (CLI).
- Practice more than 10 hours a week.
- Know what an IP address, port, and the different network protocols are.
- Know how computers on a network communicate with each other.

Learning Deliverables

1. Establish experience using LaTeX to create formatted reports
2. Create penetration testing report template
3. Perform basic reconnaissance by identifying live hosts/services, collecting evidence (screenshot/CLI output), then documenting
4. Demonstrate exploitation reasoning. After reconnaissance, select an approach, execute, then explain why the action worked or failed.
5. Build experience with nmap, Burp Suite, and Metasploit
6. Recognize attack vectors (open ports, CVEs, outdated firmware) and assess a severity score
7. Refine pen testing abilities and prepare for the industry
8. With a little extra work, you will be ready for the CompTia PenTest+ Exam, reach out if you are interested in this. I can help prepare you for what to expect.

Course Policies

Usage of AI

Usage of LLMs/AIs is allowed to aid in learning. **Use at your own discretion.** It is important to learn how to properly rely on AIs. Too much reliance will lead to it being an intellectual crutch. Don't rob yourself of learning valuable and foundational skills. Do not use AI on quizzes.

Policies on Incomplete Grades and Late Assignments

There are no deadlines in this course. All work is to be completed at your own pace by the end of the term. The course is spread out, and there should be plenty of time to complete each section. That said, you should take the extra time and dive deeper into each topic.

Academic Integrity and Honesty

Students are required to comply with the college policy on academic integrity found in the Code of Student Conduct.

Homework Submission

Submit all homework through email or on Discord.

Quizzes

The quizzes will be a link to a Google Form you must complete by the end of the term. The quizzes are relatively easy, so do not use AI to complete the quizzes. This should be a test of your own skills and where you need to improve/study. I will drop the lowest quiz grade at the end.

Optional Assignments

I've provided documents with additional material you can read through. Furthermore, I've added an optional assignment corresponding to the highlighted topics within the week. I recommend completing these to build your pen testing skills. More information can be found under the combined week documents. If you cannot find the information or need more clarity, reach out.

Grading Policy

This is a project-based independent study. If you do the work, you will pass. That said, don't undermine yourself. Learn as much as you can and refine your penetration testing skills to prepare yourself for working in the industry.

- 30%: Capstone Final
- 20%: Completing all sections under "Jr Penetration Tester"
- 20%: Quizzes
- 15%: Completing the "Bandit Network" Pentesting Report
- 10%: Completing the Network Services room
- 5%: LaTeX Template

Schedule and weekly learning goals

The schedule is tentative and subject to change. The learning goals below should be viewed as the key concepts you should grasp after each week. Each quiz will be on the outlined sections. A link to the online quiz will be provided. The walkthrough will build experience using cybersecurity pen testing tools. There will be optional boxes that can be completed to further refine your pen testing and reporting skills. I recommend completing them. There will be a final capstone, which will be a simulated pen test. You will need to complete a scan, then report the findings in a LaTeX document.

Week 01 (Jan 20–Jan 26): Reporting & LaTeX

- Understand the importance of reporting
- Familiarize yourself with Overleaf and LaTeX formatting
- Recreate the provided report template in LaTeX (this template will be reused)
- **Deliverable:** Submit a PDF of your LaTeX template + Quiz 1

Week 02 (Jan 27–Feb 02): Pre-Pentest – RoE, Scope & Authorization

- Read Week 02 packet and material
- Understand Scope vs ROE
- Understand the importance of pre-test standard practices
- **Deliverable:** Complete Quiz 2

Week 03 (Feb 03–Feb 09): TryHackMe Introduction

- Complete TryHackMe Roadmap Sections 1 & 2 (onboarding/fundamentals)
- Start Section 3 (Intro to Web Hacking)

Week 04 (Feb 10–Feb 16): Web Hacking Foundations

- Complete roadmap Section 3 (Intro to Web Hacking)
- Focus on HTTP basics, authentication/session concepts, and common web attack surfaces

Week 05 (Feb 17–Feb 23): Burp Suite Block (Section 4 Part I)

- Start roadmap Section 4 (Burp Suite)
- Take your time with this section and fully understand Burp Suite
- Understand Proxying, Intercepting, Request/Response structure (Computer Networks)

Week 06 (Feb 24–Mar 02): Burp Suite Block (Section 4 Part II)

- Complete roadmap Section 4 (Burp Suite)
- **If Time Permits:** Complete Optional Assignment
- **Deliverable:** Complete Quiz 3 (Section 3 + Section 4) & Submit Check In (I)

Week 07 (Mar 03–Mar 09): Network Security & Nmap (Section 5)

- Start roadmap Section 5 (Network Security & nmap)
- Learn scan planning and how to interpret nmap output to identify exposed services
- Understand the severity of exposed services and the attack vectors a malicious actor can utilize
- Understand how you can fingerprint a computer's OS based on the network configuration

Week 08 (Mar 10–Mar 16): Vulnerability Research (Section 6)

- Complete Section 5 & 6
- **If Time Permits:** Complete Optional Assignments
- **Deliverable:** Complete Quiz 4 (Section 5 + 6) & the **Network Services** room

Week 09 (Mar 16–Mar 23): Spring Break (no graded work due)

Week 10 (Mar 24–Mar 30): Metasploit Block (Part I)

- Learn controlled exploitation, interpreting module output, and collecting safe proof
- Take time and familiarize oneself with the Metasploit commands
- Start Section 7 (Metasploit)

Week 11 (Mar 31–Apr 06): Metasploit Block (Part II)

- **If Time Permits:** Complete optional material

Week 12 (Apr 07–Apr 13): Privilege Escalation

- Intro to post-compromise enumeration and common configuration patterns
- Understand how privilege escalation changes severity and impact
- **Deliverable:** Complete Quiz 5 (Section 7 + 8) & Submit Check In (II)
- **Deliverable:** Complete the **Bandit network** and submit pentesting report

Week 13 (Apr 14–Apr 20): Capstone - Simulated Commercial Pentest

- You will be given the Scope and RoE
- You are to test the IPs outlined in the scope for vulnerabilities & report your findings

Week 14 (Apr 21–Apr 27): Capstone - Simulated Commercial Pentest

- Complete the pen test on the network
- Report the findings in your LaTeX report
- **Deliverable:** Final penetration testing report (PDF)